Step 2: Plot the points for the above ordered pairs $(x,\ y)$ on the graph using suitable scale.

Step 3: Draw the parabola and mark the co-ordinates of the parabola which intersect the X axis.

Step 4: The roots of the equation are the x coordinates of the intersecting points $(-4,\ 0)$ and $(3,0)$of the parabola with the X axis which are -4 and 3 respectively.

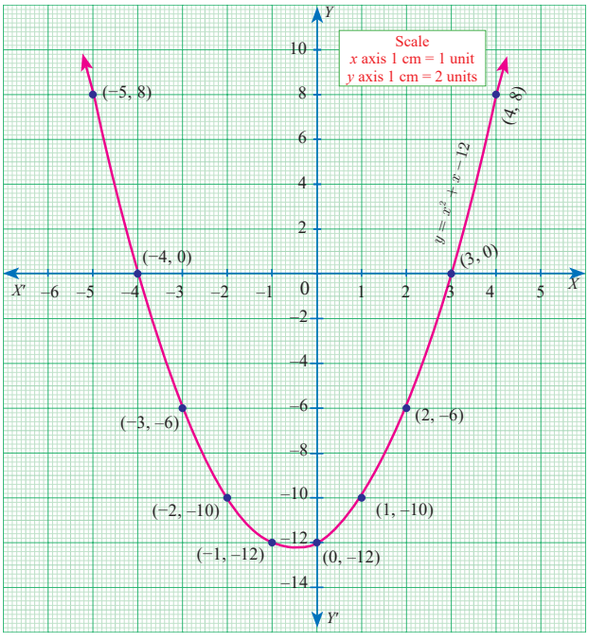


Fig. 3.19

Since there are two points of intersection with the X axis, the quadratic equation $x^2 + x - 12 = 0$ has real and unequal roots.

(ii) $x^2 - 8x + 16 = 0$

Step 1: Prepare the table of values for the equation $y = x^2 - 8x + 16$

x	−1	0	1	2	3	4	5	6	7	8
y	25	16	9	4	1	0	1	4	9	16

Step 2: Plot the points for the above ordered pairs $(x,\ y)$ on the graph using suitable scale.

Step 3: Draw the parabola and mark the coordinates of the parabola which intersect with the X axis.

Step 4: The roots of the equation are the x coordinates of the intersecting points of the parabola with the X axis $(4,0)$ which is 4.

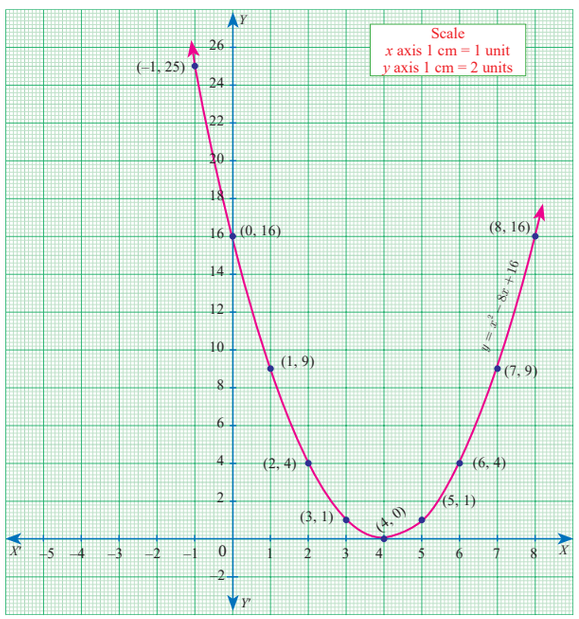


Fig. 3.20

Since there is only one point of intersection with X axis, the quadratic equation $x^2 - 8x + 16 = 0$ has real and equal roots.

(iii) $x^2 + 2x + 5 = 0$

Let $y = x^2 + 2x + 5$

Step 1: Prepare a table of values for the equation $y = x^2 + 2x + 5$

x	−3	−2	−1	0	1	2	3
y	8	5	4	5	8	13	20

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